
How Do World Action Models Use Their Futures?

Evidence from Mechanistic Interventions

Nicole Ma

Abstract

World action models jointly generate future observations and robot actions, but the functional role of their imagined futures remains unclear. Recent destructive interventions establish that these policies causally use future-position representations; they do not show what a particular coherent future tells the policy to do. We introduce reachable-donor future transplantation, which exchanges natural future representations between alternative continuations of the same physical state while fixing the current observation, instruction, action noise, and denoising schedule. The resulting signed estimand asks whether action and physical execution move toward the donor continuation. In Cosmos Policy, donor futures steer actions and executed endpoints at early states across all ten LIBERO-10 tasks; camera and content interventions favor a visible inverse-dynamics account, and effects attenuate sharply at later states. In Cosmos 3, a frozen population study over 22 saved-state clusters and six RoboLab tasks finds predicted-donor action steering of 0.992 and executed physical steering of 1.003 on a normalized recipient-to-donor scale. Token-count-preserving activation patching removes 0.877 of predicted-donor action projection. Yet isolated robot-motion and object-state pixel factors each have effects below 0.011, despite strong coherent whole-future steering. These results establish cross-generation directional future use and a mediating internal pathway, while showing that the actionable semantic carrier can be distributed or architecture dependent and does not establish general planning over task-object consequences.

1 Introduction

World action models (WAMs) couple prediction and control: a shared model generates a possible visual future and the robot action to take next [2–5, 15]. This architecture invites a planning-like interpretation. The policy appears to imagine how the scene could evolve and then act using the anticipated consequences. Joint generation alone, however, does not identify this computation. The future could be behaviorally irrelevant, could act as an undifferentiated latent workspace, or could mainly encode the robot motion already paired with the action.

RIFT substantially narrows this ambiguity [17]. Across four WAMs and all 40 LIBERO tasks, masking or reassigning future-cache values changes executed trajectories and reduces success. RIFT therefore establishes that WAM action experts causally use meaningful, position-bound future representations. It does not show what a particular coherent future tells the policy to do. Its principal intervention outcomes are deviation from the original execution and task success: they reveal dependence, but do not associate an edited future with a specific alternative behavior and ask whether the robot moves toward it.

We make that question directional. From one exactly restored physical state, two native policy branches define a recipient and donor: alternative generated futures, actions, and executed endpoints that are all reachable from the same state. We insert the donor future into the recipient computation and score the result on the signed recipient-to-donor axis. If the future merely perturbs computation,

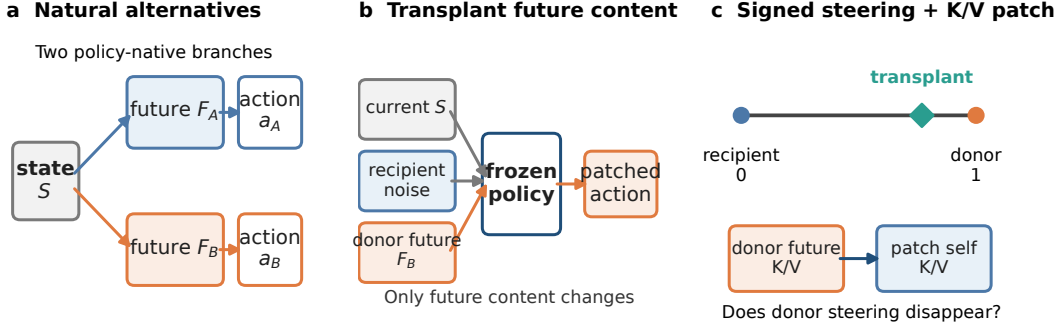


Figure 1: Reachable-donor future transplantation gives a destructive intervention a semantic direction. (a) Two policy-native continuations start at the exact same state. (b) The recipient run retains its current observation, instruction, noise, and schedule, but receives the donor future. (c) signed projections test movement toward that donor; activation patching then tests whether future-token K/V content mediates the effect.

behavior may change without aligning to the donor. If its content controls action, the patched action and endpoint should move toward the particular behavior associated with the donor. Multiple donors further test whether the intervention selects the correct alternative.

This distinction can be stated compactly: *RIFT establishes future dependence; we test semantic directionality*. Figure 1 summarizes the design. We combine representation transplantation with modality and object/robot content interventions, destructive attention localization, and token-count-preserving activation patching of future-token keys and values. This yields three linked questions: (i) can coherent future content directionally select behavior, (ii) what content carries that effect, and (iii) through which internal interface does it act?

We find strong but bounded future use. At the first decision state in Cosmos Policy, donor futures steer action and physical execution in all ten LIBERO-10 tasks. Wrist imagery carries most of the effect, primary-camera imagery carries a smaller effect, and future proprioception is null. The effect falls sharply at later registered states. A separate 20-state 2×2 intervention changes decoded robot and object futures with high target specificity but produces essentially no object-content effect. Rare natural robot-only pairs steer strongly; matched object-only pairs do not reliably identify their donor. In the released Cosmos 3 policy, predicted and physically executed donor futures steer action and robot execution across 22 independent clusters and all six eligible RoboLab tasks. Replacing donor-run future-token K/V with self-future K/V suppresses donor steering and physical execution in every cluster. A separate masked 2×2 study, however, finds that neither isolated robot pixels nor isolated object pixels reproduce a meaningful fraction of the whole-future effect.

The result is mechanistic at the representation and pathway level, not a complete circuit account. Both policies use coherent imagined visual futures to steer action, but the content carrier differs: Cosmos Policy evidence favors prospective visible robot motion, whereas Cosmos 3 requires a more compositional or distributed account. We do not find evidence that either model generally evaluates task-object consequences, searches over plans, or selects successful futures.

2 Related work

World action models. Joint image/action denoising and predictive visual policies learn futures alongside controls [2, 3, 5]. Cosmos Policy fine-tunes a pretrained video model for future prediction and visuomotor control [4]; Cosmos 3 provides a later omnimodal architecture and released DROID policy [9]. These architectures establish that future and action can be generated together, not what downstream computation uses the future for.

Recent efficiency work exposes the same ambiguity. Fast-WAM shows that explicit test-time rollout can be removed while retaining competitive performance [16]; Faster-WAM retains efficient future conditioning for robustness [19]. AGRA argues that visually plausible foresight need not be action-

grounded and modifies training to improve representation alignment [11]. These results distinguish the benefits of future-prediction training from the use of a particular generated future at inference.

Closest comparison: RIFT. RIFT records and edits the action-independent future-position K/V interface [17]. Masking, norm-matched noise, frozen-present values, and spatial or temporal re-assignment cause large trajectory drift and success loss. Replaying one final-clean cache nearly preserves execution in supported architectures, establishing consumption-side sufficiency of a fixed interface. Our question begins after that result. RIFT asks whether meaningful future values and their positions are required; we ask whether a coherent alternative future selects its associated alternative action and physical endpoint. Its EE-ADE is an unsigned deviation from the original trajectory. Our donor projection is signed by an independently generated alternative.

Mechanistic interventions. Causal mediation distinguishes information that is encoded from information actually used downstream [10, 13]. Causal tracing and path patching localize behavior by replacing internal activations or computational routes [1, 7]. Activation-patching methodology motivates clean/corrupted pairing, explicit metrics, no-op controls, and held-out validation [18]. Because patched components can interact with unpatched mediators, we interpret layer and group effects as bounded pathway evidence rather than additive circuit decomposition [12].

3 Experimental logic

3.1 Reachable-donor transplantation

Let state S admit two native branches A and B . Their generated action chunks are a_A, a_B , their future representations are F_A, F_B , and physical execution yields endpoints e_A, e_B . We rerun recipient A while holding fixed S , the language instruction, recipient action noise, future-noise path, solver, and denoising schedule. Only the clean future target changes from F_A to F_B .

For an intervened action $\hat{a}$, normalized donor projection is

$$\pi_a(\hat{a}; A, B) = \frac{(\hat{a} - a_A)^\top (a_B - a_A)}{\|a_B - a_A\|_2^2}. \quad (1)$$

Thus native recipient A is 0 and native donor B is 1; negative values move away from B , and values above 1 overshoot it. We define π_e analogously for registered physical endpoints. The principal contrast is donor clamp minus self clamp, averaged over reciprocal branch directions and future-noise draws within saved state. This signed estimand is the central difference from generic disruption.

Pairs are selected using only native action and endpoint separation, never intervention outcomes. Controls include an exact self clamp, an exact norm- and recipient-distance-matched Gaussian target, a prespecified natural control, a within-task shuffled future, and bitwise no-op recomputations. We separately clamp wrist video, primary video, and future proprioception. Every saved action condition is executed from the exact simulator state.

3.2 What content is used?

The initial transplant contains several correlated signals: robot pose, object state, camera motion, and potentially task progress. We therefore use two complementary designs. A natural-pair screen searches for robot-divergent/object-matched and object-divergent/robot-matched futures from a common recipient. These are policy-native but structurally rare. A separately frozen 2×2 study re-renders object state $O \in \{0, 1\}$ crossed with robot state $R \in \{0, 1\}$ at 20 held-out LIBERO states. This achieves broad coverage and exact factor preservation, but its hybrid videos may be off the native future manifold. We report both rather than treating one as definitive.

3.3 Pathway interventions

In Predict2-based Cosmos Policy, action queries can attend to future keys. We remove future keys at a prespecified late transformer block with a continuous gate, alongside all-key identity, random-key, current-key, and early-block controls. This is a necessity/localization test and changes attention normalization.

Table 1: Each intervention answers a different causal question. The donor experiment adds a signed semantic target that destructive tests do not provide.

Intervention	Primary readout	Supported conclusion
Mask/noise/shuffle	trajectory drift; success	future-interface dependence
Reachable donor	signed action and endpoint projection	content-specific directional sufficiency
Modality/ 2×2 content	donor projection under isolated content	functional content used by the policy
Self-K/V patch into donor run	loss of donor-directed projection	pathway-level content mediation

Cosmos 3 permits a cleaner content-mediation test. We first record future-video keys and values from a self-future run. During a donor-future run, we replace only future-token K/V at selected layers; token count, token order, current-video K/V, text K/V, action K/V, current observation, instruction, and noise remain unchanged. Record/replay and self-patch arms must be bitwise exact. If donor projection falls after the patch, the replaced future content mediates that donor-directed effect. We report the absolute reduction in donor projection (“mediation loss”), not a population natural indirect effect.

3.4 Statistical inference

The independent unit is an exactly restored simulator state. Reciprocal directions, multiple donors, noise draws, and process repeats are repeated measurements. Unless stated otherwise, intervals are 10,000-resample cluster bootstraps over saved states; task-balanced sensitivities first average states within task and then bootstrap tasks. We preregistered 0.10 donor-projection units—ten percent of the native recipient-to-donor displacement—as the smallest effect of interest for the Predict2 confirmatory family and the Cosmos 3 population and factor families. Registered primary-family sign tests receive Holm correction.

4 Do alternative futures directionally steer actions?

4.1 Cosmos Policy: population-level test

We first study the released Cosmos-Policy-LIBERO-Predict2-2B checkpoint [4] on LIBERO-10 [6]. The frozen grid contains 30 unseen units: ten tasks crossed with a first query, a state after three open-loop chunks, and a state after six chunks. Eight native branches determine one reciprocal donor pair per state. Three future-noise draws are repeated within direction. Physical endpoints concatenate goal-relevant simulator coordinates with the official nine-dimensional proprioceptive endpoint. Pair selection, controls, attention block, outcome construction, and bootstrap are frozen before intervention outcomes.

At the first query, donor-minus-self action steering is 0.499 (95% CI [0.336, 0.679]), positive in all ten tasks. Executed endpoint steering is 0.552 ([0.387, 0.728]), also positive in all ten. The action contrast remains positive against Gaussian (0.497), prespecified natural (0.103), and shuffled-future (0.255) controls. Physical contrasts are respectively 0.549, 0.110, and 0.277. This is not merely an action-vector change: the simulator endpoint moves in the independently defined donor direction.

The effect is state dependent (Figure 2a). After three chunks, mean action steering falls to 0.031 and physical steering to 0.004. After six chunks they are 0.027 and 0.025. Across all 30 registered states, action steering is 0.186 ([0.095, 0.290]) and physical steering is 0.194 ([0.091, 0.305]), but the lower intervals narrowly miss the 0.10 practical threshold. Because each timing stratum uses a different initial-state index, this is a state/timing association rather than a pure causal effect of elapsed time. The supported claim is strong steering at identifiable early states, not uniform future mediation throughout an episode.

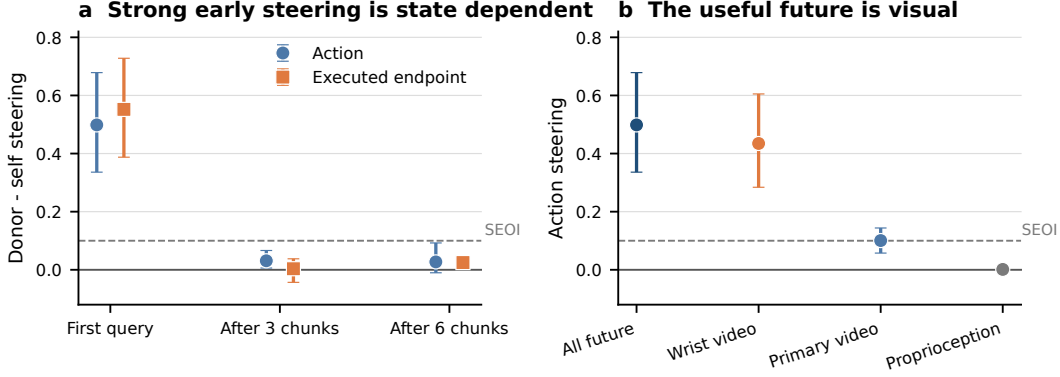


Figure 2: Predict2-based Cosmos Policy uses visual futures strongly at early states. Points are saved-state means and bars are 95% cluster-bootstrap intervals. (a) action and executed endpoint steering are large at the first query and sharply attenuate at later registered states. (b) wrist video carries most first-query action steering, primary video carries a smaller effect, and future proprioception is null. SEOI: preregistered smallest effect of interest (0.10).

4.2 Cosmos 3: cross-generation replication

We port the same reachable-donor logic to the released 16B Cosmos3-Nano-Policy-DR0ID checkpoint [9] and public RoboLab tasks [14]. Cosmos 3 uses 36 two-way full-attention layers, four UniPC denoising calls, 33 raw video frames mapped to nine VAE latent frames, and 32 absolute joint-position actions. The intervention targets only unconditioned future-video coordinates at the final guided sampler boundary. Identity layout capture, identity sampler wrapping, cache record/replay, and self patches reproduce actions bitwise; sentinels verify zero direct mutation of action coordinates. Each physical condition reconstructs the environment, restores the same initial state, and replays the prefix exactly.

Before inspecting confirmatory interventions, a frozen native census selected 24 saved-state clusters across six eligible tasks, capped at four per task, using only native success, action and endpoint separation, and replay validity. Twenty-two clusters completed. Two registered native branches terminated before producing the full 33-frame videos required as intervention sources; these units remain missing and were not replaced. The prespecified minimum of 20 clusters across five tasks was therefore achieved. The independent unit is saved state; reciprocal directions and source types are repeated measures.

Predicted-donor action steering is 0.992 (95% state-bootstrap CI [0.972, 1.010]), positive in 22/22 clusters and every task. It remains 0.433 ([0.332, 0.538]) above a preselected natural-future control. For futures encoded from physically executed reachable branches, donor-minus-executed-self action steering is 0.983 ([0.890, 1.064]), and executed physical endpoint steering is 1.003 ([0.950, 1.050]), both positive in 22/22 clusters. Executed-donor action steering also exceeds the geometry-matched Gaussian by 0.941 ([0.837, 1.032]) and the natural control by 0.391 ([0.246, 0.533]). Task-balanced bootstraps agree with every conclusion. All registered primary donor and K/V sign tests remain positive after Holm correction.

The four-task multi-donor engineering study additionally verifies correct-alternative specificity: action identifies the correct donor in 12/12 cases for predicted and executed sources. The population result now establishes cross-task generality in a later, substantially different policy rather than only repeated engineering feasibility. It still does not identify mediation of task success: no screened branch pair in either policy retained a robust binary success/failure label across continuation seeds.

5 What information in the future causes the steering?

5.1 Visual modalities

Modality clamps localize the early result to vision (Figure 2b). Wrist-future action steering is 0.435 ([0.284, 0.605]), primary-camera steering is 0.101 ([0.057, 0.144]), and future-proprioception steer-

Table 2: Main signed effects organized by scientific question. Intervals cluster on saved state.

Study	Units	Effect (95% CI)	Main reading
Predict2 first query	10 states	action: 0.499 [0.336, 0.679]	coherent donor selects action
Predict2 first query	10 states	endpoint: 0.552 [0.387, 0.728]	direction survives execution
Cosmos 3 population	22 states, 6 tasks	action: 0.992 [0.972, 1.010]	all predicted donors steer
Cosmos 3 population	22 states, 6 tasks	physical: 1.003 [0.950, 1.050]	direction survives execution
Predict2 visual content	10 states	wrist: 0.435 [0.284, 0.605]	visual robot motion dominates
Predict2 2×2 object	20 states	action: 0.00015 [-0.00012, 0.00049]	isolated object content is null
Natural robot-only	4 states	action: 0.291 [0.060, 0.521]	strong but exploratory
Cosmos 3	22 states, 6 tasks	loss: 0.877 [0.842, 0.910]	future K/V mediates direction
predicted-future K/V patch			
Cosmos 3 pixel factors	10 states, 6 tasks	robot: 0.0022; object: 0.0061	neither isolated factor is meaningful

ing is 0.001 ($[-0.004, 0.008]$). Because the wrist view tightly tracks the manipulator, this pattern predicts a prospective visible-motion or inverse-dynamics-like role.

5.2 Robot-motion versus object-state content

The content studies support that interpretation while exposing its limits. In the 20-state 2×2 study, decoded futures identify the intended target cell with 92.5% primary-camera and 98.75% wrist-camera top-1 accuracy. Nevertheless, the object main effect is 0.00015 on action similarity ($[-0.00012, 0.00049]$) and 0.00018 on executed goal-state steering ($[-0.00040, 0.00102]$), far below 0.10. Endpoint target-cell identification is exactly chance (0.25). The manipulation changes the decoded object future, but execution does not follow it.

Natural matching avoids rendered hybrids but yields only four eligible units, below the frozen minimum of ten. In those units, robot-only future transplantation gives action steering 0.291 ($[0.060, 0.521]$) and executed robot-endpoint steering 0.315 ($[0.059, 0.571]$). Object-only effects are much weaker: action 0.021 and goal endpoint 0.021 with an interval crossing zero; correct-donor top-1 is at each unit’s chance rate. We therefore do not infer that object consequences are universally unused. We infer that the observed steering is best explained by prospective visible robot motion, while meaningful isolated object-state use is unsupported in these experiments.

This content conclusion is deliberately narrower than the directional result. Natural object-only alternatives are rare, whereas rendered 2×2 hybrids may lie off the policy’s native future manifold. Donor videos also contain robot pose and correlated visual changes. “Prospective visible robot motion” is therefore the best-supported functional description at the studied Cosmos Policy states, not a fully decoded latent variable or a universal claim that WAMs ignore object consequences.

Cosmos 3 provides an important counterpoint. A separately frozen factor family selects 12 population units round-robin by native object-position separation; ten complete across all six tasks, meeting its prespecified minimum. Starting from the native recipient video, an object-priority disjoint mask copies donor object pixels, donor robot pixels outside the object mask, both, or neither. The isolated robot main effect is 0.0022 on action ($[-0.0016, 0.0060]$) and 0.0102 on the full executed endpoint ($[-0.0067, 0.0275]$). The isolated object effect is 0.0061 on action ($[0.0009, 0.0119]$) and -0.0001 on the full endpoint ($[-0.0254, 0.0239]$). Every interval lies inside the preregistered ± 0.10 equivalence region, and task-balanced estimates agree. The detectable object action effect is only 0.6% of native donor displacement and is practically negligible. Small-denominator object-specific endpoint ratios are unstable and are not used for equivalence claims.

Thus strong whole-future steering does not reduce to either isolated pixel factor in Cosmos 3. This cross-model difference rules out a universal robot-pixel account: actionable content may require coherent robot-object context, be distributed outside the masks, or interact nonlinearly with tokenization. The study does not show that robot and object factors are jointly necessary.

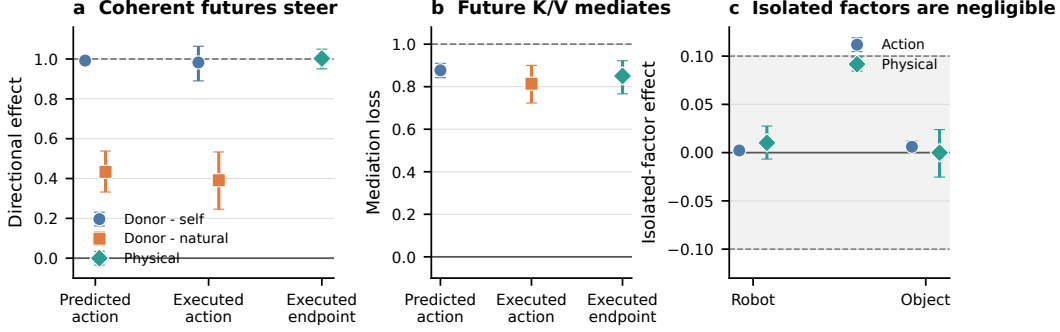


Figure 3: Cosmos 3 population results. Points and bars are saved-state means and 95% cluster-bootstrap intervals. (a) coherent predicted and executed futures steer action, beat natural controls, and move physical endpoints toward the donor across 22 clusters and six tasks. (b) token-count-preserving self-future K/V replacement removes most donor steering. (c) in ten factor units across six tasks, isolated robot and object pixels have effects within the preregistered ± 0.10 equivalence region.

6 Through what internal pathway does the future affect action?

6.1 Cosmos Policy: an active future-to-action attention edge

Removing future keys from action queries at block 27 yields a monotonic action-disruption curve: normalized L2 is 0.022, 0.035, 0.054, and 0.068 at gates 0.25, 0.50, 0.75, and 1.00. The all-key re-computation is bitwise exact. Full removal is more disruptive than equal-count random-key removal by 0.016 and than block-0 future removal by 0.048. Equal-count current-key removal is larger by 0.024. Executed endpoints show the same bounded conclusion: the late future edge is causally active, but neither the only nor the largest route through the block.

6.2 Cosmos 3: future-token K/V content mediates steering

Layer and pathway choice are frozen from excluded calibration tasks, where direct-interface patching removes 0.842 projection units ([0.766, 0.910]) and exact record/replay and self-patch gates pass. In the confirmatory population, predicted-future K/V replacement removes 0.877 action-projection units ([0.842, 0.910]), positive in 22/22 clusters. Executed-future replacement removes 0.814 action units ([0.723, 0.900]) and 0.851 physical-endpoint units ([0.766, 0.922]), again positive in every cluster. Task-balanced estimates are 0.877, 0.812, and 0.853, with intervals excluding zero and the 0.10 smallest effect of interest. All primary family sign tests survive Holm correction.

Together, the two models provide complementary pathway evidence rather than identical circuit analyses. Cosmos Policy shows that a late future-to-action attention edge is necessary enough to measurably affect behavior. Cosmos 3 preserves token count and attention normalization while replacing future-token content, giving stronger evidence that donor-directed information is mediated through future-token K/V. Layerwise and grouped patches can still interact with unpatched mediators [12]; they identify an active internal pathway, not a complete or uniquely necessary circuit.

7 Discussion

What is the imagined future for? The shared account across models is prospective visual action conditioning: a particular coherent future specifies a corresponding action direction, and the policy reads this relation through future-token content. Cosmos Policy further resembles visually mediated inverse dynamics: the strongest modality is the wrist camera, natural robot-only donors steer strongly, and future proprioception and isolated object-state content do not [3]. Cosmos 3 does not admit the same isolated-pixel explanation. Its whole future steers almost completely while robot-only and object-only edits are each negligible, suggesting distributed content, contextual binding, or tokenizer-mediated interactions.

This interpretation is stronger than saying that future prediction is a useful auxiliary loss, and more specific than saying that corruption hurts. It is weaker than planning. We do not observe search over possible futures, comparison of values, stable success/failure mediation, or meaningful isolated control by predicted object outcomes. “Imagined” describes the generated representation; it does not imply deliberation.

Relationship to RIFT. The papers answer adjacent questions. RIFT shows that the future interface is read and that its values and positions are necessary for normal execution [17]. We show that a coherent donor future supplies a signed control signal: action and execution move toward the behavior paired with that donor. Content isolation then asks what that signal represents, and activation patching asks where donor-directed influence is mediated. The correct contrast is dependence versus semantic directionality, not performance versus steering.

Broader impact. Mechanistic tests can make embodied policies more auditable by separating plausible-looking predictions from representations that actually control behavior. The same steering interfaces could also be misused to redirect robot actions or could create false confidence if validated only on narrow simulators. Deployment should therefore require state-distribution coverage, physical safety constraints, and monitoring beyond latent intervention success.

8 Conclusion

World action models do not merely react when their future cache is damaged. At identifiable states, the semantic content of a coherent imagined future controls which action they take. Reachable-donor transplantation makes that claim directional, and token-count-preserving K/V patching localizes donor steering to the future-token pathway across a frozen Cosmos 3 population. Content interventions reveal an informative model difference: Cosmos Policy favors prospective visible robot motion, whereas Cosmos 3’s effect is not reproduced by isolated robot or object pixels. The resulting account is mechanistic but bounded: these policies use coherent imagined futures to steer robot motion, without current evidence that they generally plan over task-object consequences.

References

- [1] Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching, 2023. URL <https://arxiv.org/abs/2304.05969>. arXiv preprint, version 2.
- [2] Yanjiang Guo, Yucheng Hu, Jianke Zhang, Yen-Jen Wang, Xiaoyu Chen, Chaochao Lu, and Jianyu Chen. Prediction with action: Visual policy learning via joint denoising process. In *Advances in Neural Information Processing Systems*, volume 37, pages 112386–112410. Curran Associates, Inc., 2024. doi: 10.52202/079017-3570. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/cbe25fa0e7c7084049276888a09acc8d-Abstract-Conference.html.
- [3] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 24328–24346. PMLR, 2025. URL <https://proceedings.mlr.press/v267/hu25g.html>.
- [4] Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos Policy: Fine-tuning video models for visuomotor control and planning, 2026. URL <https://arxiv.org/abs/2601.16163>. arXiv preprint.
- [5] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model, 2025. URL <https://arxiv.org/abs/2503.00200>. arXiv preprint, version 3.
- [6] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 44776–44791. Curran Associates, Inc., 2023. doi: 10.52202/075280-1939. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c3c666820ea055a77726d66fc7d447f-Abstract-Datasets_and_Benchmarks.html.
- [7] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. doi: 10.52202/

- 068431-1262. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.
- [8] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-scale simulation of household tasks for generalist robots. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024. doi: 10.15607/RSS.2024.XX.050. URL <https://www.roboticsproceedings.org/rss20/p050.html>.
- [9] NVIDIA. Cosmos 3: Omnimodal world models for physical AI. *arXiv preprint arXiv:2606.02800*, 2026. doi: 10.48550/arXiv.2606.02800. URL <https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf>.
- [10] Judea Pearl. Direct and indirect effects. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence*, pages 411–420, San Francisco, CA, 2001. Morgan Kaufmann. URL https://ftp.cs.ucla.edu/pub/stat_ser/R273.pdf.
- [11] Lu Qiu, Yizhuo Li, Yi Chen, Yuying Ge, Yixiao Ge, and Xihui Liu. Making foresight actionable: Repurposing representation alignment in world action models, 2026. URL <https://arxiv.org/abs/2606.12217>. arXiv preprint.
- [12] Sankaran Vaidyanathan, David Arbour, Aaron Mueller, Scott Niekum, and David Jensen. The curse of multiple mediators: Hidden interaction effects in activation patching, 2026. URL <https://arxiv.org/abs/2606.27510>. arXiv preprint.
- [13] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc., 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html>.
- [14] Xuning Yang, Rishit Dagli, Alex Zook, Hugo Hadfield, Ankit Goyal, Stan Birchfield, Fabio Ramos, and Jonathan Tremblay. RoboLab: A high-fidelity simulation benchmark for analysis of task generalist policies. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026. URL <https://arxiv.org/abs/2604.09860>.
- [15] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, Ayaan Malik, Kyungmin Lee, William Liang, Nadun Ranawaka, Jiasheng Gu, Yinzheng Xu, Guanzhi Wang, Fengyuan Hu, Avnish Narayan, Johan Bjorck, Jing Wang, Gwanghyun Kim, Dantong Niu, Ruijie Zheng, Yuqi Xie, Jimmy Wu, Qi Wang, Ryan Julian, Danfei Xu, Yilun Du, Yevgen Chebotar, Scott Reed, Jan Kautz, Yuke Zhu, Linxi “Jim” Fan, and Joel Jang. World action models are zero-shot policies, 2026. URL <https://arxiv.org/abs/2602.15922>. arXiv preprint.
- [16] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination?, 2026. URL <https://arxiv.org/abs/2603.16666>. arXiv preprint, version 2.
- [17] Chushan Zhang, Jinguang Tong, Xuesong Li, Yikai Wang, and Hongdong Li. Keep the future, drop the rollout: RIFT for world action models, 2026. URL <https://arxiv.org/abs/2608.11521>. arXiv preprint, version 2.
- [18] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=HF17y6u9BC>.
- [19] Weiheng Zhao, Haoyi Jiang, Xin Shi, Liu Liu, Fan Huang, Zhizhong Su, Wei Sui, and Xinggang Wang. Faster-WAM: Efficient inference-time future conditioning for robust world action models, 2026. URL <https://arxiv.org/abs/2608.04404>. arXiv preprint.

A Reproducibility details

The Predict2 study uses public Cosmos Policy commit 18a2acca, checkpoint revision cb689ec0, the public LIBERO benchmark, PyTorch 2.7/CUDA 12.8, and two NVIDIA H100 80GB GPUs. Its frozen seeds, state grid, pair-selection rule, controls, intervention coordinates, and 10,000-resample bootstrap are stored in the accompanying configuration. The Cosmos 3 study pins

356 public Cosmos commit 9aa98e5a, Cosmos Framework commit d4599e2e, RoboLab com-
357 mit 9db0aaf0, checkpoint revision 2b9f9517, container identifiers, all branch seeds, the 36-
358 layer token census, and exact identity gates. The checked-in frozen manifest has SHA-256
359 46b6fe3d6abb07712d0926361587b853b2e8313a10c7faeecb0150d030e986b. Full per-run
360 timing and aggregate compute accounting remain to be added before submission.

361 **B Additional result boundaries**

362 The six-state RoboCasa replication [8] is exploratory and is not pooled with LIBERO. Its donor-
363 minus-self physical endpoint effect is 0.0287 (95% CI [0.0101, 0.0492]), positive in 6/6 states; ac-
364 tion steering is smaller and does not reliably beat the Gaussian control. The Predict2 continuation
365 screen yields no robust binary success/failure pair after registered replication seeds. In the Cosmos
366 3 population, two of 24 frozen units are missing because native intervention-source branches termi-
367 nated before producing the required 33-frame videos (after 21 and 29 actions); neither was replaced.
368 The prespecified 20-unit/five-task population minimum and 10-unit/five-task factor minimum are
369 both achieved across six tasks.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Sections 1, 4–7 state the directional steering claim, its state dependence, and the boundary against planning or success mediation.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Sections 4–6 discuss limitations where each result is interpreted: model and state coverage, timing confounding, object-factorization distribution shift, pathway interactions, and the absent binary-success estimand.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper reports empirical intervention results and does not claim a theorem or formal proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sections 3–6 and Appendix A specify the experimental units, estimands, controls, clustering, bootstrap, public checkpoints, pinned revisions, seeds, and exactness gates needed to reproduce the main findings.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The intervention code and machine-readable results are organized for release, but an anonymous public artifact is not yet attached to this manuscript draft.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sections 3–6 provide the state grid, branch construction, intervention targets, controls, action horizons, model interfaces, and statistical analysis; exact revisions and seeds are summarized in Appendix A.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: All central means have explicitly defined 95% cluster-bootstrap intervals, the independent clustering unit and 10,000-resample procedure are stated in Section 3.4, and exact sign counts are reported where central.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Appendix A reports GPU type and software environment, but complete per-run wall-clock and aggregate compute-hour accounting is still being assembled for the submission artifact.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The experiments use public robot-policy checkpoints and simulation benchmarks, involve no people or personal data, and report negative and failed intervention results rather than selectively discarding them.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 7 discusses auditability benefits, action-redirection misuse, false confidence from narrow simulation, and corresponding validation and safety-monitoring needs.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: This work does not release a new high-risk pretrained model or scraped dataset; it analyzes existing robot policies in simulation.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Existing assets are cited and exact source revisions are recorded, but an explicit license inventory for every upstream code and benchmark asset remains to be added before submission.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [No]

Justification: New intervention code and result manifests are documented in the working artifact, but the anonymized release package and its final license are not yet attached to this draft.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The research contains no crowdsourcing and no human participants.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

686 Justification: The research contains no human-subject data collection or interaction.

687 Guidelines:

- 688 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
- 689 with human subjects.
- 690 • Depending on the country in which research is conducted, IRB approval (or equiva-
- 691 lent) may be required for any human subjects research. If you obtained IRB approval,
- 692 you should clearly state this in the paper.
- 693 • We recognize that the procedures for this may vary significantly between institutions
- 694 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
- 695 guidelines for their institution.
- 696 • For initial submissions, do not include any information that would break anonymity
- 697 (if applicable), such as the institution conducting the review.

698 **16. Declaration of LLM usage**

699 Question: Does the paper describe the usage of LLMs if it is an important, original, or

700 non-standard component of the core methods in this research? Note that if the LLM is used

701 only for writing, editing, or formatting purposes and does *not* impact the core methodology,

702 scientific rigor, or originality of the research, declaration is not required.

703 Answer: [N/A]

704 Justification: Language models were used only for writing and formatting assistance, not

705 as a component of the experimental method or scientific evaluation.

706 Guidelines:

- 707 • The answer [N/A] means that the core method development in this research does not
- 708 involve LLMs as any important, original, or non-standard components.
- 709 • Please refer to our LLM policy in the NeurIPS handbook for what should or should
- 710 not be described.